

COMPANY A · BUSINESS PROPOSAL · CONFIDENTIAL

Predict-and-Save:

An AI-Driven Retention Engine for India's Wireless Telecom Market

Turning 100,000 customer records into ₹3.79 bn of annual net benefit

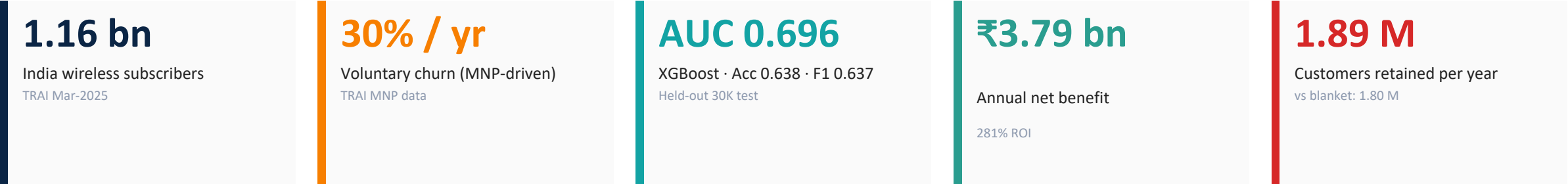
Kaileshwar R K

Executive Summary

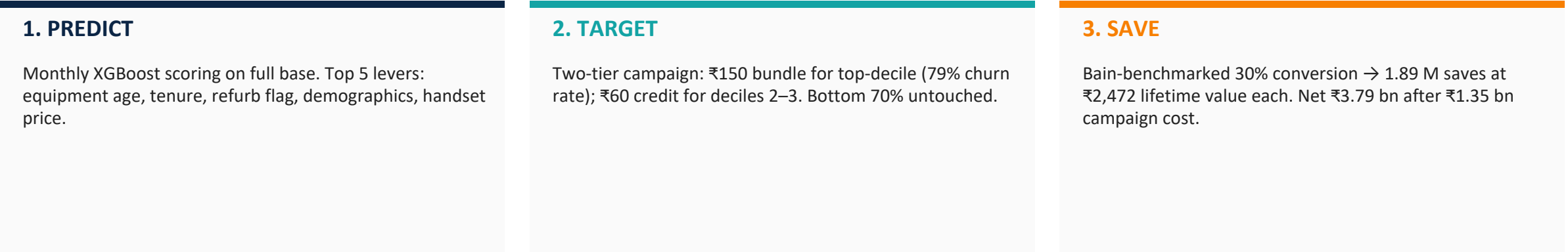
Onepage.The whole proposal in five numbers.

The proposal in one sentence

Use machine learning to score Company A’s 50 M-subscriber base monthly, then deploy two tiered retention offers against the top 30% — recovering ₹3.79 bn in annual net benefit at 281% ROI.



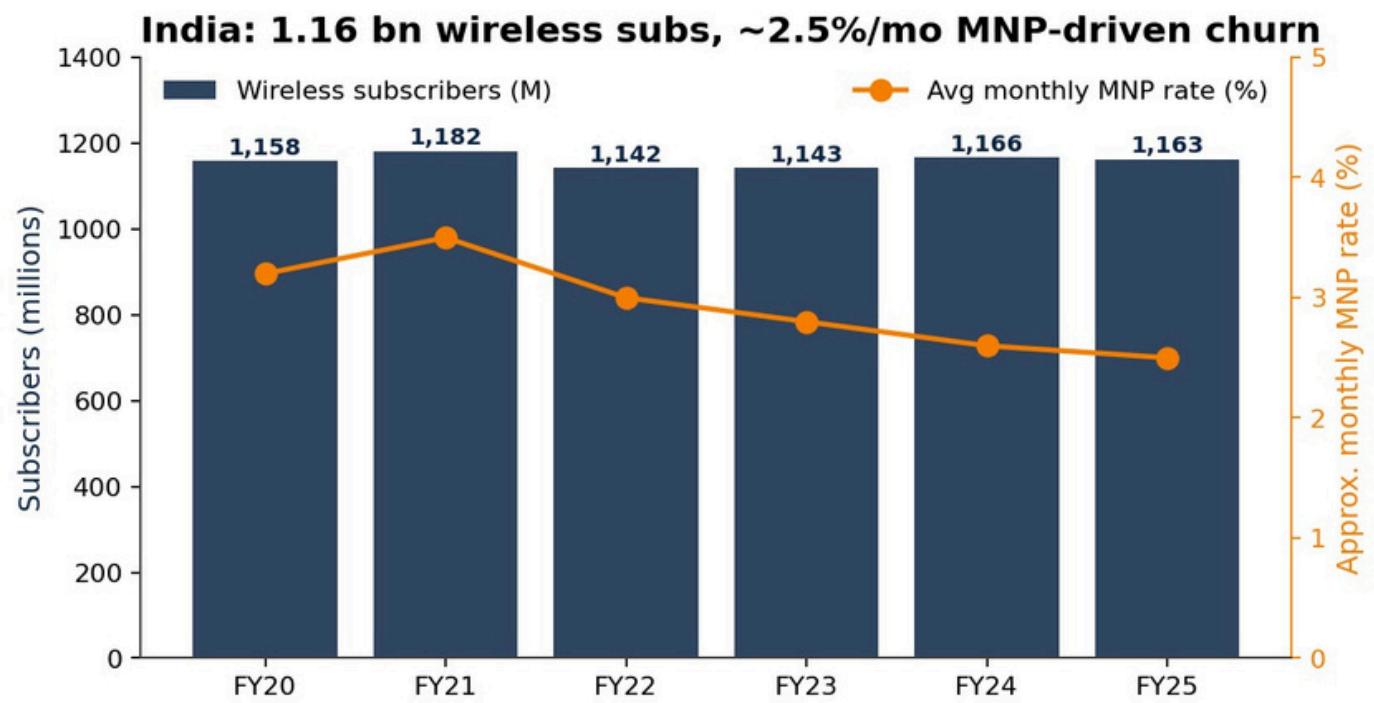
Three pillars of the proposal



Sources: TRAI Performance Indicator Report (Mar 2025); Reliance Jio Q4 FY25; Bain & Co. Telecom Loyalty 2023.

India's Telecom Market: Why Churn Is The Battle

Mature, 1.16 bn subs market where MNP makes switching frictionless



Three structural facts

1.16 bn subs

World's 2nd-largest wireless market; 96% mobile-broadband penetration.

~30% / yr churn

TRAI logs 7-8 M MNP requests every month — voluntary, frictionless porting.

ARPU ₹206

Q4 FY25 record (Reliance Jio); 24-mo gross-profit CLV ≈ ₹2,472.

CAC > 3× retention

Acquiring a replacement subscriber costs 3-7× retention spend (Bain).

Saturated market

Growth = stealing from competitors or lifting ARPU per existing customer.

Dataset: 100,000 Customers, ~100 Features, 1 Question

`Client.csv` + `Record.csv` joined on Customer_ID

Client.csv — demographics

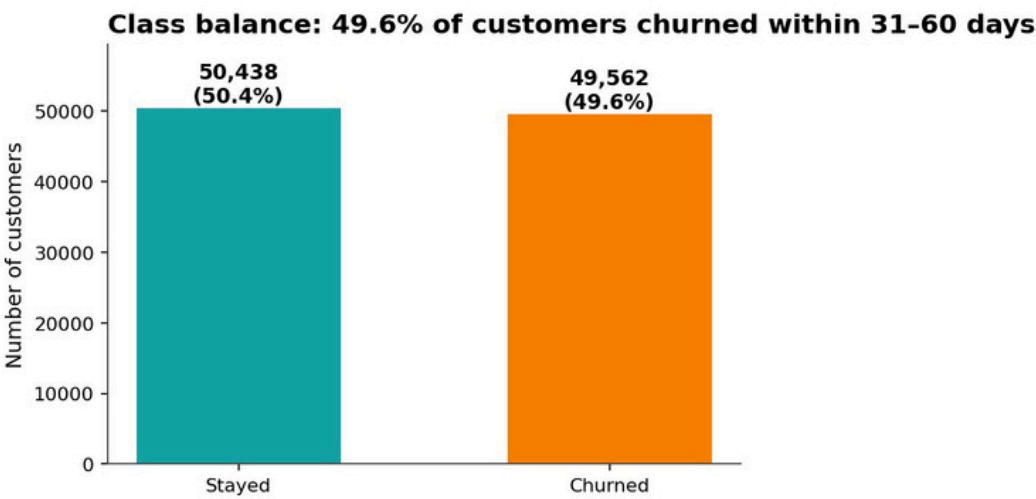
- 100,000 rows · 50 columns
- Tenure (months), credit class, handset price
- Household: income, dwelling, marital, kids
- Geography: area, ZIP-derived demographics
- Equipment: age, refurb/new, web-cap

Record.csv — usage

- 100,000 rows · 51 columns
- Mean monthly revenue, MOU, charges
- Quality: dropped/blocked/unanswered calls
- Customer-care call volumes
- Trend: change_mou, change_rev
- Target: churn (31–60d binary flag)

After merge

- Shape: 100,000 × 100
- 21 categorical · 79 numeric
- Some demographic fields >35% NaN
- Target balance: 49.6% churn / 50.4% stay
- 70/30 stratified train-test split

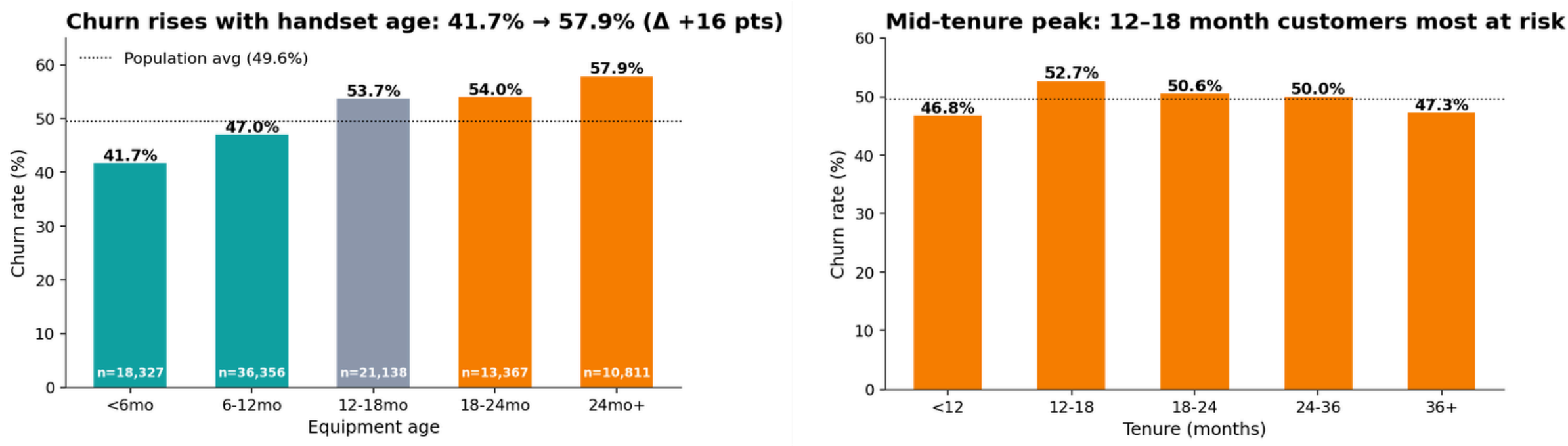


Missingness — handled in modelling

numbcars	49.4%	Tree handles missing natively
dwllsize	38.3%	Tree handles missing natively
HHstatin	37.9%	Tree handles missing natively
ownrent	33.7%	Tree handles missing natively
income	25.4%	Tree handles missing natively

EDA: Two Levers The Business Can Actually Pull

Equipment age and tenure show the cleanest churn signals — both intervenable

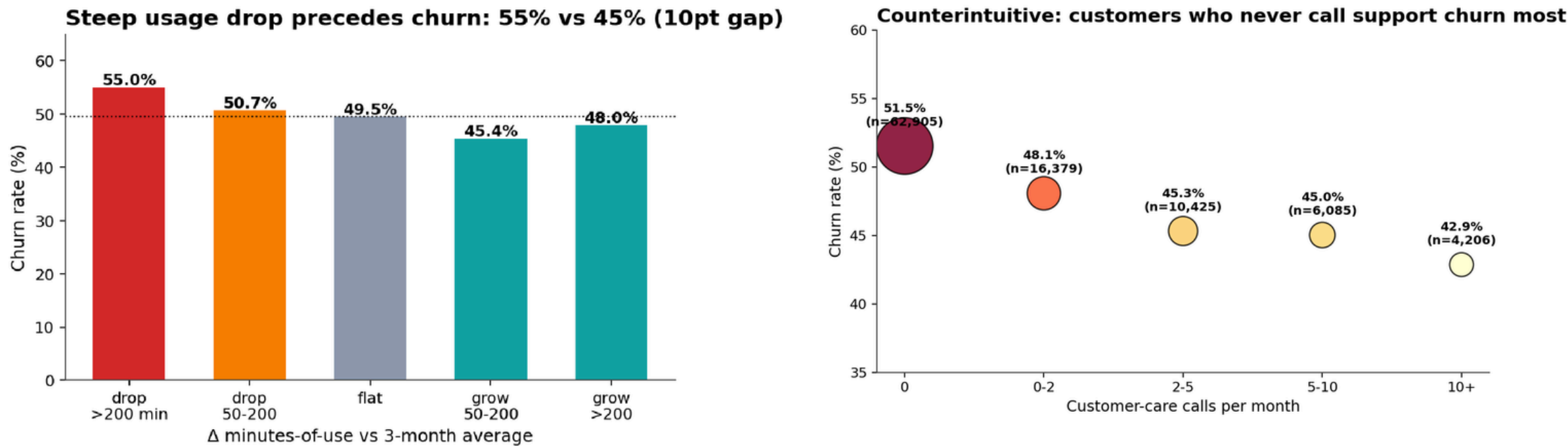


What this means for the proposal

- Customers with 24+ month-old handsets churn at 57.9% — 16 points above new-handset customers (41.7%). A scheduled handset-upgrade nudge is a natural intervention.
- Mid-tenure (12–18 months) is the danger zone: contract anniversary effect. Pre-anniversary outreach materially shifts the curve.

EDA: The Quiet Customers Are The Dangerous Ones

Two behavioural signals that contradict intuition



Two non-obvious patterns

- Usage drop predicts churn: a >200-min/mo decline lifts churn from 49.6% baseline to 55.0%. This is a real-time signal — billing systems already capture it.
- Customers who never call support churn most (51.5%). Silent dissatisfaction beats vocal complaint. Proactive outreach to silent customers is an under-exploited lever.

The business question, the ML question, and the metric that connects them

Business question

Which Company A customers will leave in the next 31–60 days, and how much profit can we save by acting on the prediction?

ML framing

Binary classification on `churn`. Balanced classes (49.6%/50.4%) → accuracy and F1 both meaningful; AUC stays comparable across models.

Decision output

Per-customer risk score (0–1). Sorting and bucketing into deciles is what a retention team needs — not a yes/no label.

Why we chose churn (and not revenue or usage)

Churn (chosen)

Direct revenue impact (CLV at risk). Action is binary (offer / no offer). Tutorial-recommended starting point.

Revenue regression

Useful but lacks an action. Predicting "how much" doesn't tell us what to do — and the proposal must recommend an action.

Usage forecasting

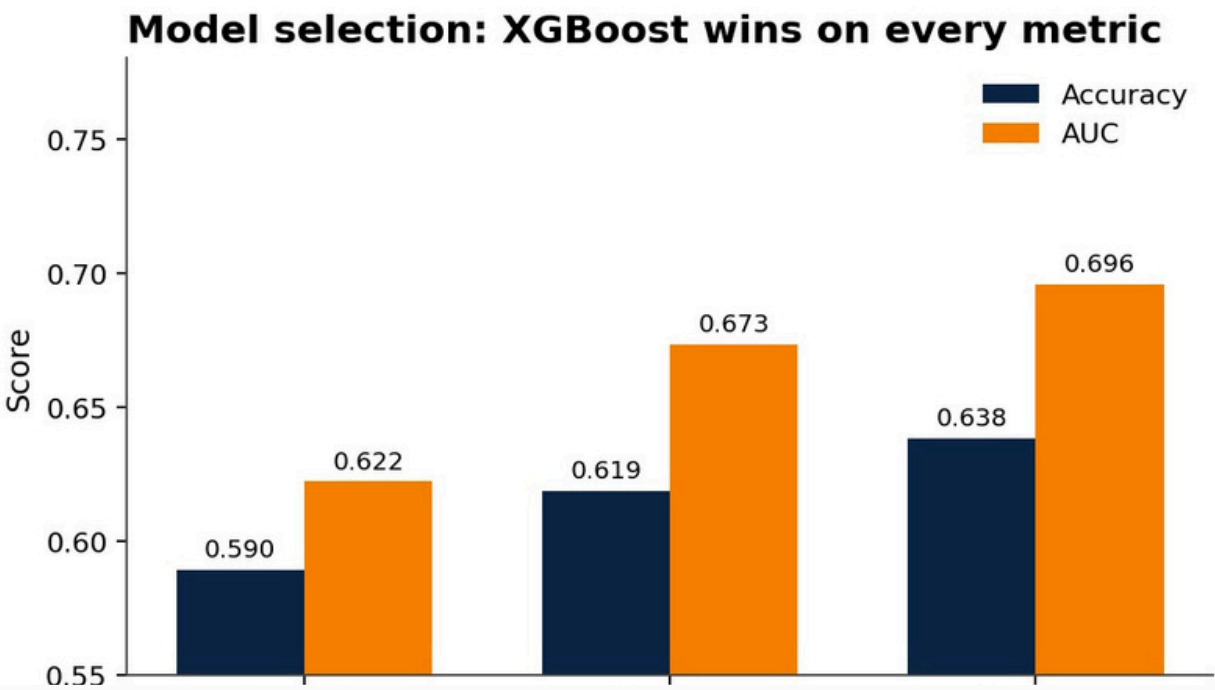
Indirect: would still need a downstream rule to convert into intervention.

Segmentation only

Descriptive, not predictive. We use it inside the proposal (as the tier-A/B carve-up) but it can't stand alone.

Model: XGBoost — Why It Wins

A three-model bake-off; XGBoost beats both baselines on every metric



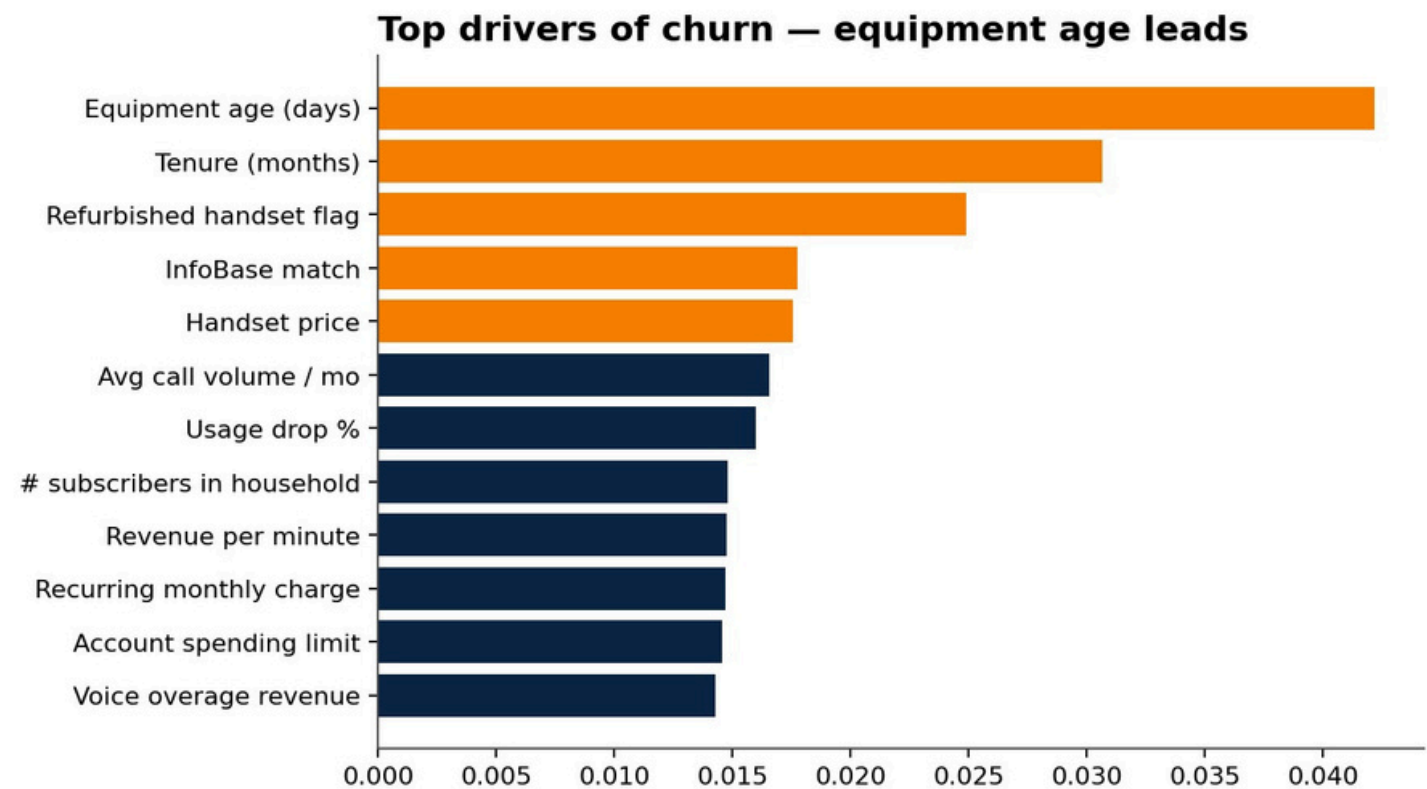
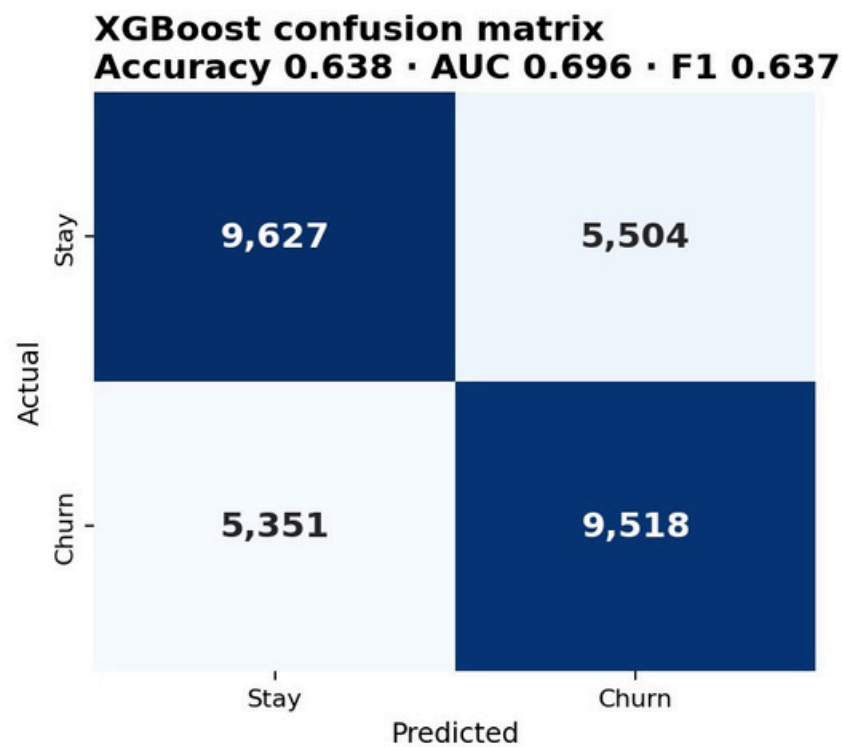
Chosen model: XGBoost	
500 trees · depth 6 · LR 0.05 · 70/30 stratified	
Accuracy	AUC
0.6382	0.6958
F1 score	PR-AUC
0.6368	0.6838
Precision	Recall
0.6336	0.6401

Why XGBoost — three reasons

- Native NaN handling — 5 demographic features have >25% missing values; XGBoost learns the missingness pattern itself.
- Tabular gradient boosting is the best-in-class baseline for ~100K-row mixed-scale datasets (no feature scaling needed).
- Sklearn-compatible API + interpretable feature importance (gain) → easy to deploy and easy to explain to a non-ML executive.

What The Model Sees: Confusion Matrix & Top Drivers

Where it gets right vs wrong, and which signals it uses



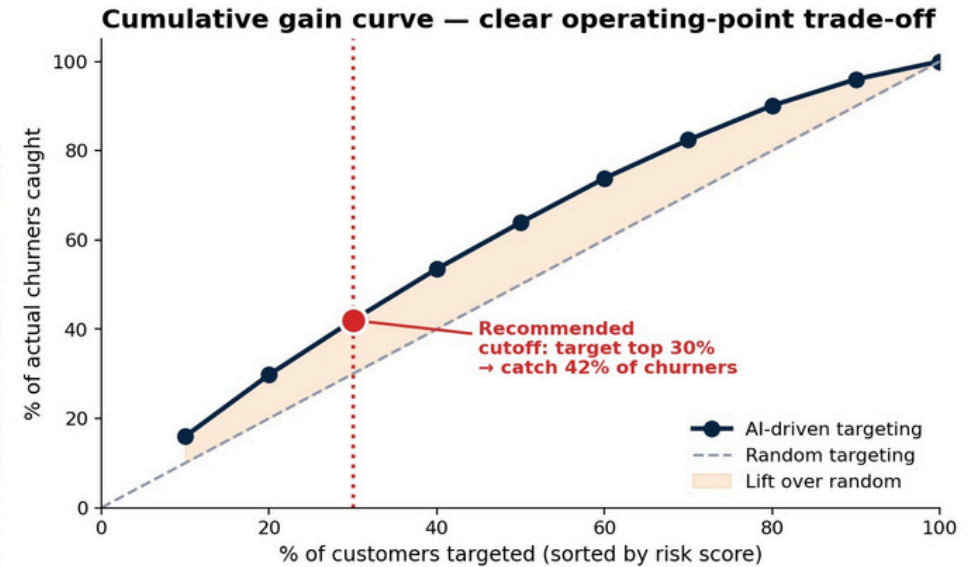
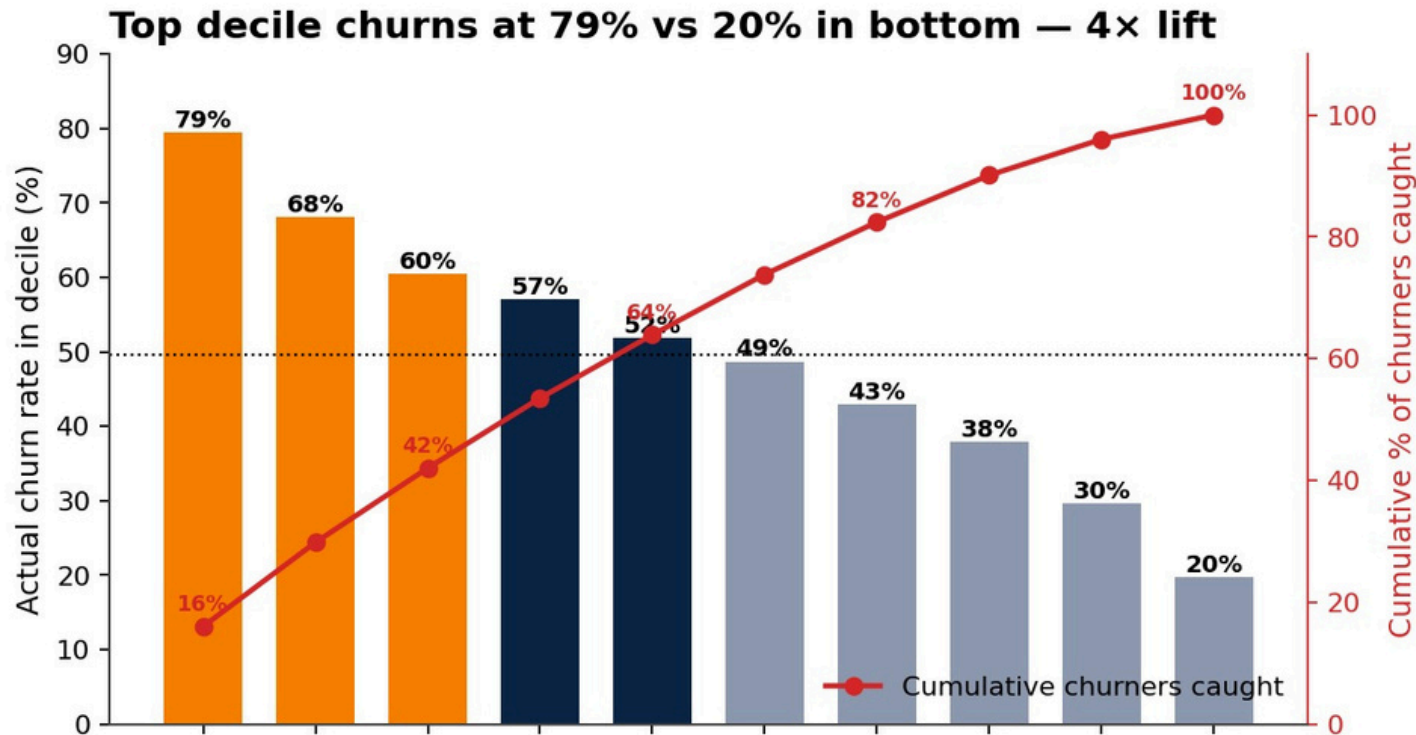
From driver to lever — the actionable five

Equipment age, tenure, refurb-flag, demographics-match, handset-price are the top-5 by gain. Three are intervenable today — handset upgrade campaigns, anniversary offers, refurb-replacement programmes — and form the tier-A/B carve-up in slide 11.

Source: Author analysis (XGBoost gain importances on 70K-row train fold, validated on 30K test).
Company A · AI-Driven Retention Proposal · Kailesh · GCI 2026

From Probability To Action: The Decile Lift

Top-10% customers churn 4× more than bottom-10% — the campaign cutoff falls out of this curve



Operating point: target the top 30%

- Decile 1 alone catches 16% of churners; the top three deciles together catch 42%.
- Targeting the top 30% by predicted risk is 1.4× more efficient than random outreach (purple-shaded lift area).
- The cutoff is a business choice, not an ML one — every percentage point of capture costs more campaign rupees.

Our Recommendation: A Two-Tier Retention Engine

Different offers for different risk — bottom 70% never receive a discount

TIER A — Top 10% 5.0Mcustomers · 79%churn rate Offer ₹150 bundle (2 GB extra/mo + ₹100 bill credit) Logic Where 16% of all churners live. High-touch: in-app message + outbound call.	TIER B — Decile 2-3 10.0Mcustomers · 53%churn rate Offer ₹60 bill credit (SMS-driven activation) Logic Where another 26% of churners live. Self-service offer; auto-applies on bill.	NO ACTION — Bottom 70% 35.0Mcustomers · 38%churnrate Offer — (holdout group) Logic Cheaper to lose at the margin than to discount everyone. Holdout enables A/B measurement.
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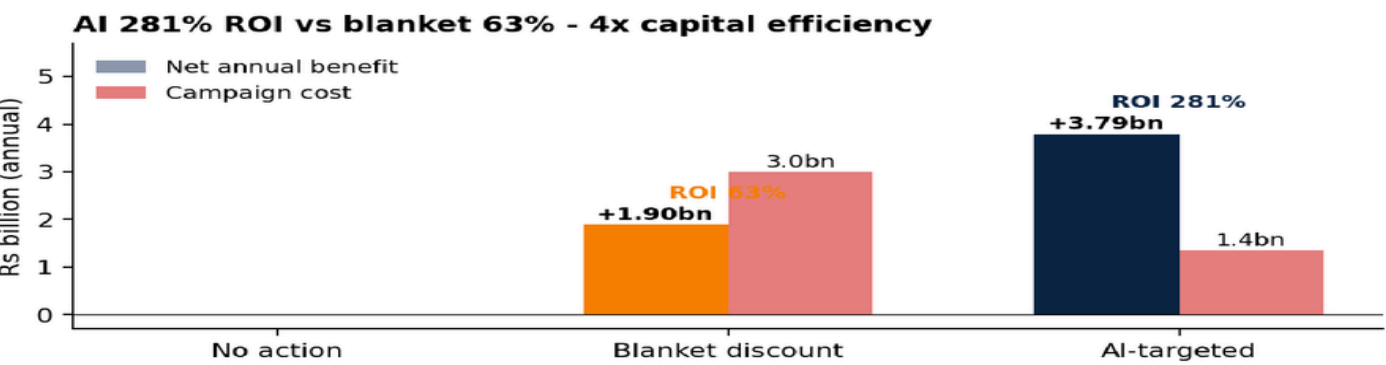
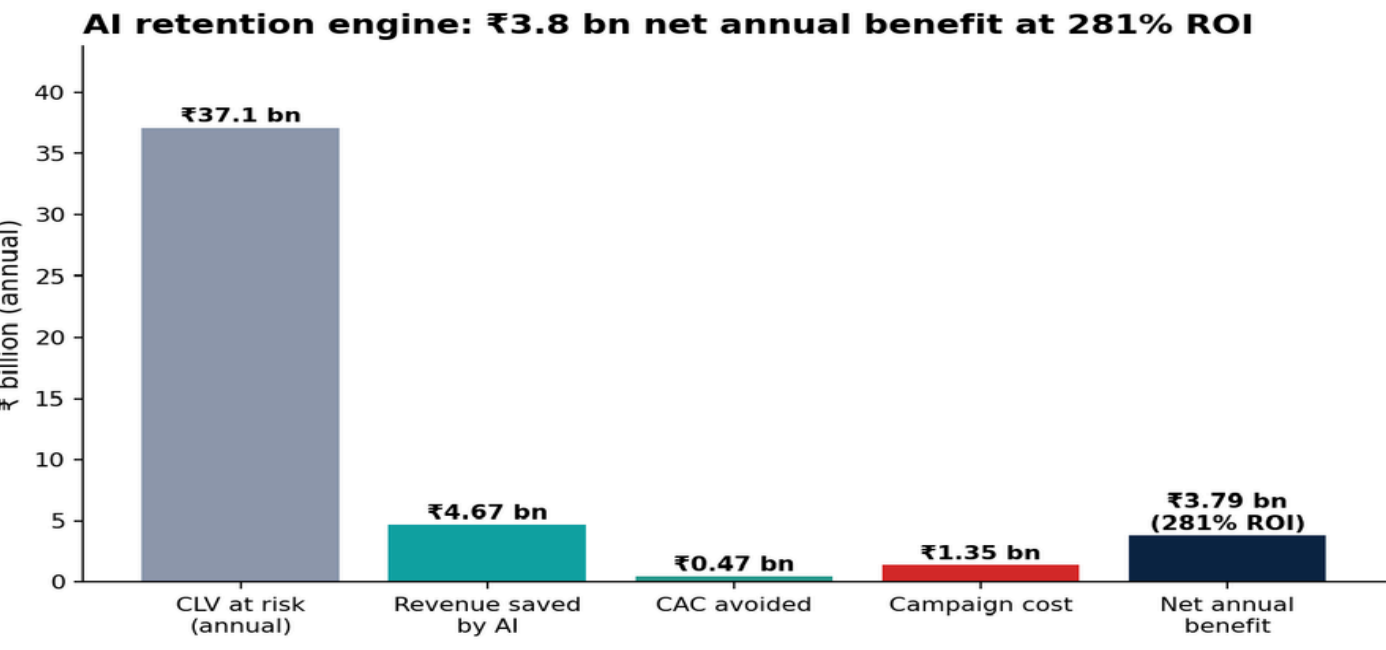
Operational playbook · monthly closed-loop retention



Offer values benchmarked against Bharti Airtel & Vi retention promo schemes (FY24 disclosures); A/B holdout follows GroupM telecom-CRM playbook.
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Quantified Impact: ₹3.79 bn Net Annual Benefit

Every assumption sourced. Sensitivity on the next slide.



The math, line by line

Operator base	50 M subs
Annual churn	30% → 15 M
CLV (24-mo gross)	₹2,472 / sub
AI captures	42% → 6.30 M
Conversion (Bain)	30% → 1.89 M
Revenue saved	₹4.67 bn
CAC avoided	₹0.47 bn
Campaign cost	₹1.35 bn
Net benefit	₹3.79 bn
ROI	281%

AI vs blanket

4.5× higher capital efficiency:
281% ROI (AI) vs 63% (blanket).
Blanket discounts subsidise the safe 70% — AI engine concentrates spend where it changes outcomes.

Will The Numbers Hold? Sensitivity & Risks

The proposal stays profitable across a wide band of assumptions

Net annual benefit (₹ bn) — sensitivity grid

Conv. rate ↓ / CLV horizon →	12 mo	18 mo	24 mo
15% (worst)	₹0.4	₹1.3	₹2.3
20%	₹0.8	₹2.1	₹3.3
25%	₹1.3	₹2.8	₹4.4
30% (base)	₹1.7	₹3.6	₹5.5
35% (best)	₹2.1	₹4.3	₹6.5

Even at half-Bain conversion (15%) and half horizon (12 mo), the engine still nets ₹0.4 bn at 31% ROI.

Top 4 risks & mitigations

- Model decay**

Quarterly retraining; champion-challenger A/B before promotion.
- Offer leakage**

A/B holdout (10% of decile 1-3) measures incrementality vs natural retention.
- Cannibalisation**

Tier-A offer is one-time; no auto-renewal. Caps revenue dilution at <2% ARPU.
- Regulatory (DPDP Act 2023)**

No demographic targeting outside aggregate segments; data on India-cloud only.

Implementation Roadmap & Pricing

90-daypilot · 6-monthrollout · ongoingmanaged-servicecontract

90-day pilot timeline

Days 1-30 Discovery & Data	Days 31-60 Pilot Build	Days 61-90 Live Pilot	Months 4-6 Scale
- Onboard data feeds - Validate features - Refresh model on 6-mo data	- Deploy scoring API - A/B design + holdout - Tier-A offer setup	5 M customer pilot - Daily KPIs - Go/no-go decision	- Roll out to full base - Tier-B activation - Quarterly retrain cadence

Pricing — fixed pilot fee + outcome-linked managed service

Pilot fee	₹2.5 cr	90-day fixed-scope build & validate
Annual licence	₹3.0 cr	Model hosting, scoring, monthly refresh
Performance fee	5% of net benefit	Capped at ₹15 cr/yr (vs ₹3.79 bn benefit)
Total Year-1	~₹24.5 cr	<7% of projected ₹3.79 bn net benefit

Pricing model patterned on standard Indian IT-consulting MSAs (Big-4 / domestic Tier-1); performance fee structure aligns engagement incentives.
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Everyexternalclaimcited;fullURLsinOmnicampusreferences field

Market context (slides 2, 3)

TRAI	Performance Indicator Report (Quarterly), Mar 2025. https://traigov.in/
Reliance Jio	Q4 FY25 Investor Presentation. ARPU ₹206/mo disclosure. https://www.ril.com/ar2024-25/
Bharti Airtel	FY24 Annual Report — retention scheme disclosures. https://www.airtel.in/about-bharti/equity/results
GSMA	Mobile Economy India 2024. https://www.gsma.com/mobileeconomy/india/

Method (slides 7, 8, 9)

Chen & Guestrin	XGBoost: A Scalable Tree Boosting System (KDD 2016). https://arxiv.org/abs/1603.02754
scikit-learn	Pedregosa et al., JMLR 12 (2011). https://scikit-learn.org/
Tutorial reference	GCI World 2026 final-assignment tutorial.ipynb (provided).

Business case (slides 11, 12, 13)

Bain & Co.	The Loyalty Effect (Reichheld, 1996; 2023 update). https://www.bain.com/insights/
Reichheld, F.	Zero Defections: Quality Comes to Services. HBR 1990.
Vodafone Idea (Vi)	FY24 retention scheme — bundle / bill-credit benchmarks. https://www.myvi.in/
Govt. of India	Digital Personal Data Protection Act 2023. MeitY official text.

Tools

Python 3.9 · pandas · NumPy · matplotlib · seaborn · scikit-learn · XGBoost · python-pptx

Generative AI	Claude (Anthropic) — used for slide formatting, diagram drafting, and copy editing. Chat URL not shareable externally due to internal security policies; happy to walk through the workflow on a call.
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